Fiscal Year:	FY 2006	**Task Last Updated:**	FY 02/12/2007
PI Name:	Lackner, James R. Ph.D.		
Project Title:	Coordination of Turning and Reaching		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02454-9110	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-07
Start Date:	12/02/2002	**End Date:**	12/02/2006
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 12/2/2006 per J. Dardano (4/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	DiZio, Paul (Brandeis University)		
Grant/Contract No.:	NAG9-1466		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We have developed a turn and reach paradigm to assess post-flight disturbances of arm-trunk coordination. Our recent ground-based work has shown that simultaneous turn and reach movements involve high rotational velocities of the trunk that generate large, potentially disrupting Coriolis forces on the reaching arm and on balance. Nevertheless, reaching accuracy is preserved because the central nervous system programs compensation for the self-generated Coriolis forces. Failure to predict accurately Coriolis force magnitude produces disruption of coordination and potentially loss of balance. Our aims are: Aim 1-Parametrically evaluate Coriolis force influences on kinematics and dynamics of voluntary turning and reaching to targets under conditions of normal and increased body weight. Aim 2-Measure reaching movements and postural control after motor adaptation to rearrangement of the relationship between voluntary torso rotation relative to the feet and inertial torso rotation in space. Aim 3-Assess whether motor adaptation to Coriolis force perturbed reaching acquired in a rotating room transfers to turn and reach movements in a natural environment. These		

	experiments will be of value in pinpointing factors contributing to post-flight postural and arm-trunk coordination deficits. They will also have significance for understanding potentially contributing factors in populations at risk for falling.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Coriolis forces on the arm produced by torso rotation during everyday activities are much larger than interaction torques between arm segments. Several patient populations, including stroke patients and Parkinson's patients have been shown to suffer from specific deficits in control of interaction torques among limb segments. The large Coriolis torques on the limb due to torso rotation may perturb both the limb and overall posture. Other work in our laboratory is investigating the possibility that labyrinthine-loss patients lacking semicircular canal inputs may have impaired coordination of trunk and limbs during voluntary turning and pointing movements.
Task Progress:	We have had three specific aims and we have successfully completed all of them. Aim 1. Evaluate parametrically Coriolis force influences on kinematics and dynamics of voluntary turning and reaching to targets under conditions of normal and increased body weight. We have systematically evaluated the influence of Coriolis forces on reaching movements in ground based laboratory experiments and parabolic flight. The experimental results demonstrate that under 1g conditions subjects when turning and reaching for a target are very accurate, regardless whether they reach slow or fast, or without visual feedback. Normal subjects show temporally overlapping arm and torso velocities, with the peak velocities occurring virtually simultaneously. As a consequence, very large Coriolis forces are generated on the reaching arm, nevertheless these forces do not disrupt movement accuracy. By contrast, reaching movements during passive body rotation have large trajectory deviations and endpoint errors. These results indicate that in natural turn and reach movements there is an anticipatory feed forward compensation for the impending Coriolis force which maintains straight and accurate reaches in external space coordinates. Subjects exposed to 1.8g force levels in parabolic flight initially exhibit disruptions of movement paths and endpoints. Within 40-50 turn and reach movements they achieve a great deal of compensation. Aim 2. Measure reaching movements after adaptation to movement-contingent rotary motion of the base of support. These experiments have been completed. Subjects who are exposed to incremental positive or negative gain changes in the rotation of the surface of support linked to their voluntary trunk rotation adapt to those changes without being aware that the base of support is moving. When the platform gain is suddenly reduced to zero, the subjects show large errors in turning behavior reflecting the persistence of adaptation to the now absent platform motion. Aim 3. Measure transfer of adaptation from rotating room to turn and reach movements. Adaptation to Coriolis force perturbations of reaching movements in the rotating environment is very rapid, with most adaptation being completed in 10-20 movements. This adaptation persists post-rotation and disrupts the path and endpoint of post-rotation movements. The adaptation is manifest whether the subjects are seated or standing and with trunk stationary or voluntarily turning. We have developed a mathematical model that allows us to compute the inverse dynamics of turn and reach movements, movements made in a rotating reference frame, and movements perturbed by a mechanical manipulandum. It also enables us to parcellate the forces.
Bibliography Type:	Description: (Last Updated: 03/31/2017)
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Articles in Peer-reviewed Journals	Hudson T, Lackner JR, DiZio P. "Rapid adaptation of torso pointing movements to perturbations of the base of support." Exp Brain Res. 2005 Sep;165(3):283-93. PMID: 15942737 , Sep-2005
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